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Jaeyun (Jay) Ree

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LinkedIn,GitHub,Website

Final-year Software Engineering (Honours) student at Monash University, graduating Nov 2026, with research interests in quantum and classical algorithms for combinatorial optimisation. As a Research Assistant under Dr Mohammed Eunus Ali, I designed and benchmarked matching and assignment algorithms for autonomous-vehicle coordination, resulting in a short paper currently under review at ACM SIGSPATIAL 2026. I also completed a Human Research Ethics-approved Honours thesis on human-centred adaptive assessment, and bring software engineering experience as co-founder and CTO of AIVA. I am a recipient of the 2023 and 2024 Monash Engineering Dean's Honours.

Publications & Research Output

Under review

- J. Ree, M. E. Ali, and M. A. Cheema, "Dynamic Platoon Formation: Spatio-Temporal Matching for Towing-Based Cooperative Autonomous Mobility" (short paper), submitted to the 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL 2026). *Under review.*

In preparation

- Manuscript in preparation extending my Honours thesis on human-centred adaptive assessment (comparing LLM-based and rule-based adaptive questionnaires), targeting submission in 2026.

Research Experience

Research Assistant, Monash University, Clayton Victoria **Nov 2025 -**
Project: Dynamic Platoon Formation of Multi-Type Autonomous Vehicles for Sustainable Urban Mobility

- Contributed to modelling the coordination task as a constrained optimisation problem with multi-segment matching and point-wise capacity constraints, under Dr Mohammed Eunus Ali (with Prof. Aamir Cheema as co-author).
- Designed and implemented two matching algorithms, a Greedy Maximum-Weight Matching baseline and an Iterative Linear Assignment method using the Jonker-Volgenant solver, achieving 25-52% energy-proportional distance coverage across synthetic scenarios with up to 400 passive and 80 active vehicles.
- Built an end-to-end experimental framework in **Python** (NumPy, SciPy, Pandas) with parameter-sweep experiments and comparative benchmarking; revised and resubmitted the work across two conference formats in response to peer review.

Final Year Project Researcher (Honours Thesis), Monash University **Jul 2025 -**
Human-Centred Adaptive Assessment (FIT4701 / FIT4702)

- Conducted a human-centred study comparing rule-based and LLM-based adaptive mental-health questionnaires, supervised by Dr ABM Russel with co-supervision from Prof. John Grundy (ARC Laureate Fellow) and industry mentorship from Mark Foley (Future Wellness Group).
- Designed and ran a MUHREC-approved within-subjects study (Project ID 51664) with 40 participants, grounded in the COSMIN content-validity framework, and co-developed "Navigator", a Flutter/Supabase prototype implementing both adaptation methods.
- Led the statistical analysis (paired-sample t-tests, effect sizes) and authored the final report; the LLM-based method scored significantly higher on all three dimensions (Clarity, Relevance, Quality). Continuing the work toward a manuscript for submission in 2026.

Professional Experience

Teaching Associate, Monash University, Melbourne Victoria **Mar 2026 - Jun 2026**

- Delivered weekly applied sessions for *FIT1045: Introduction to Programming*, mentoring students on C++ fundamentals and the SplashKit library, and assessing portfolio submissions against structured rubrics.

Chief Technology Officer (CTO), AIVA, Melbourne **Dec 2025 -**

- Co-founded and led the end-to-end architecture of a full-stack AI executive-assistant platform (Node.js/Express and Supabase/PostgreSQL backend, Flutter iOS client, and an autonomous Google Gemini agent), directing technical strategy and code review across the team.

Full Stack Developer Intern, Job Search Works, Melbourne **May 2025 - Nov 2025**

- Built core features for an AI-powered job-application SaaS (resume-job matching, automated workflows) using **React**, **TypeScript**, and **Firebase**; set up CI/CD with **GitHub Actions** and Husky.

IT Analyst, Hardware Co-Ordination Intern, Coles Group **Jul 2024 - Dec 2024**

- Coordinated hardware lifecycle management across Coles divisions; automated a manual reporting process with VBA (4 hours to 30 minutes) and contributed to over $90,000 in documented operational savings through improved vendor and lifecycle planning.

Education

Monash University **July 2020 - Present [Expected Graduation: Nov 2026]**

- Bachelor of Engineering (Honours) in Software Engineering; GPA 3.759 / 4.000, WAM 82.571 (High Distinction).
- Relevant coursework in algorithms and data structures, discrete mathematics, databases, and operating systems. Proficient in **Python** (NumPy, SciPy, Pandas), **Java**, and **TypeScript/JavaScript**; experienced with Git, Linux, and SQL.

Monash College (Foundation Year) **July 2019 - June 2020**
GPA 4.0/4.0 (High Distinction)

Awards & Honours

2024 Monash Engineering Dean's Honours (Jun 2025)
2023 Monash Engineering Dean's Honours (May 2024)
Summer Vacation Research Scholarship, Faculty of IT (2025)
Industry-Based Learning Placement Scholarship (Jul 2024)
Academic Commendation for Highest Grade in FIT2081: Mobile Application Development (May 2024)
Engineering International Undergraduate Excellence Scholarship (Jun 2020)

Certifications

AWS Certified Cloud Practitioner (Mar 2025)

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